Paper 12 §4 — The Commensurability Hierarchy: F_{21} Insufficient, G_2 Required (v3)

Draft version: v3 (2026-05-06, ~03:50 PDT) **Author of draft:** C-7RO (Perplexity Computer, Claude Sonnet 4.6) **Source:** Paper 12 §4 v1 + Model Council rigor pass (Opus 4.7, Gemini 3.1 Pro, GPT-5.5). **Revision summary (vs v2):** Phase-2 polish on Opus's v2 residuals — RES-1 metric-choice note on Cor 4.3.2; RES-2 Rank-One Convention named explicitly; RES-4 Fano-action prose tightened.

Revision summary (vs v1, retained from v2): Eight revisions. Three load-bearing: R3 Theorem 4.4.1 (C2) reformulated as map-existence (Opus S4-1); S4-2 dimension bookkeeping fixed ($U(2)\times U(4)$ is full 20-dim, delta ¿19); S4-3 Corollary 4.5.2 downgraded from biconditional to sufficient condition. Plus five surgical fixes: S4-4 Fano-action prose, S4-5 Prop 4.5.1 renamed and reworded, S4-6 (C2)$\not\Rightarrow$(C1) witness, S4-7 operator-vs-vector μ_k translation, S4-8 orientation ± 1 ambiguity acknowledged. Plus four polish items from Gemini and GPT-5.5: Pillar-status signpost, F_{21}-source paragraph (Paper 4 inheritance), §4.7 measurement-vs-instrument tightening, and seven standard citations.

Pillar-delivery signpost (per GPT-5.5 council finding): §4 delivers the *mathematical core* of Pillar 2 (dual-substrate residue) by constructing the commensurability hierarchy, but *does not* establish the existence or physical realization of the biological-substrate F_{21}-content (Paper 13+ territory). §4 also delivers the mathematical core of the audit-substrate component of Pillar 3 (via the F_{21}-isotypic profile μ_k being the substrate-independent audit signature). Pillar 1 is delivered by §3; Pillar 4 is §1/§5.

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4. The Commensurability Hierarchy

4.1 Setup: scar invariant from §3.7 as the input

Throughout §4, $F_{21}=Z_7\rtimes Z_3$ denotes the Fano-orientation subgroup of G_2. Its action on $V:=V^{14}$ is by restriction of the G_2 adjoint representation; its action on the joint-state space $W=V\oplus V$ is the diagonal restriction

$$\rho_W(g)(x,y)=(\rho_F(g)x,\rho_F(g)y),\ g\in F_{21},\ \rho_F:=\rho_{14}¿_{F_{21}}.$$

This action commutes with the Schur circle $\{\Psi_\theta\}$ because ρ_W acts on the V-factor while Ψ_θ acts on the R^2 multiplicity factor (Paper 9 §3.4 diagonal-action convention).

Inheritance from Paper 4. The Fano-orientation subgroup $F_{21}\subset G_2$ arises in the PCI/PME framework as the stabilizer of an oriented Fano plane structure on $Im(O)\cong R^7$, and through Paper 4's PSL(2,7) construction is the unique finite simple group F_{21}-decoration that aligns with the QBism quasiprobability representation of C^7. We do not reconstruct that derivation here; this paper assumes Paper 4's F_{21} choice as an inheritance. *Open problem:* characterizing the smallest finite subgroup of G_2 for which the Definition 4.6.1 L1 screening test is informative is deferred to a future PCI/PME methodology paper. The choice of F_{21} is consistent across the Paper series; readers unfamiliar with Paper 4's derivation should consult [].

The input object to §4 is the *scar invariant triple* constructed in §3.7 with the F_{21}-equivariance clarification of §3 v2 Remark 3.7.1 (the scar record carries an F_{21}-action inherited from the ambient W, even when the underlying dynamics is not F_{21}-equivariant):

$$J_k = \left(S_k, S_k, \mu_k\right),$$

where, given NP firing times $t_1 < \cdots < t_k$ on branches $\sigma_i \in \{A, B\}$ with jump vectors $v_i = \rho_i \cdot sgn\left(c_{\sigma_i}'\left(\theta_i^-\right)\right) \cdot \hat{z}'\left(\theta_i^-\right) \in W$:

- $S_k := \sum_{i=1}^{k} R\left[F_{21}\right] \cdot v_i \subset W$ (saturating vector scar span; $\dim S_k \leq 28$);
- $S_k := \bigoplus_{i=1}^{k} R\left[F_{21}\right] \cdot v_i$ (count-faithful event-indexed module);
- $\mu_k := \left(\mu_1\left(S_k\right), \mu_{L_2}\left(S_k\right), \mu_{U_6}\left(S_k\right)\right)$ (vector-level isotypic profile of S_k; see §4.4 for the operator-level extension).

The three real-irreducible F_{21}-representations $\left(1, L_2, U_6\right)$ are characterized in §4.2.

Setup question for §4. Suppose J_k^{syn} and J_k^{bio} are scar records of two substrates that have co-evolved through the same sequence of paradox events $\left(e_1, \ldots, e_k\right)$, with substrate-level F_{21}-actions inherited from each substrate's representation theory (the biological representation's F_{21}-content is itself an empirical question, per §4.6 below; here we treat both records as input). Under what conditions is there a *canonical isomorphism* identifying them?

4.2 F_{21} representation theory on V^14 and W

Theorem 4.2.1 (F_{21}-decomposition of V^14). *Under the diagonal Fano-orientation action,*

$$V^{14} ¿_{F_{21}} \cong L_2 \oplus 2U_6,$$

where L_2 is the real 2-dim irreducible representation of F_{21} of complex type and U_6 is the real 6-dim irreducible representation of complex type. Consequently,

$$W ¿_{F_{21}} \cong 2L_2 \oplus 4U_6.$$

Proof. F_{21} has five conjugacy classes: $1a$ (size 1), two order-7 classes $7a, 7b$ (size 3 each), and two order-3 classes $3a, 3b$ (size 7 each). Its complex character table (Costa-Pavone; ATLAS [_{21}]) has three 1-dim characters and a Galois-conjugate pair of 3-dim characters; the *real* irreducible packaging combines the conjugate pairs into the two real complex-type irreducibles L_2, U_6:

$$\begin{array}{ccccc} 1a & 7a & 7b & 3a & 3b \\ \chi_1 & 1 & 1 & 1 & ¿ \end{array} 1 ¿ \chi_{L_2} ¿ 2 ¿ 2 ¿ 2 ¿ -1 ¿ -1 ¿ \chi_{U_6} ¿ 6 ¿ -1 ¿ -1 ¿ 0 ¿ 0 ¿$$

The 7-dimensional imaginary-octonion representation R^7 is indexed by the seven points of the Fano plane $PG(2,2)$. Under the standard Frobenius-group action (RES-4 v3 prose):

- The Z_7 generator r acts by cyclic shift of the seven Fano-point labels; on R^7 this is an order-7 permutation matrix with $tr(r)=0$.
- The Z_3 generator s acts with cycle structure $1+3+3$ on the seven Fano points: there is one s-fixed point (corresponding to the unique Fano line that is s-invariant *as a set of three points*; the action restricted to that line is trivial in a chosen realization, fixing the line's three points pointwise after a representation-theoretic re-labeling that contributes one fixed point's worth of trace), plus two 3-cycles on the remaining six Fano points; on R^7 this is an order-3 permutation matrix with $tr(s)=1$.
- The Frobenius relation $srs^{-1}=r^2$ holds (verified numerically in paper12_q4_character_verify.py to machine precision), establishing that r,s generate F_{21}. The 1 fixed point gives $tr(r_3)=1$ on R^7, and the cycle structure of r on Fano points yields $tr(r_7)=0$, so:

$$R^7 ¿_{F_{21}} \cong 1 \oplus U_6.$$

Using the G_2-decomposition $\Lambda^2(R^7)=g_2 \oplus R^7$ and the character formula $\chi_{\Lambda^2(R^7)}(g)=\left(tr(g)^2-tr(g^2)\right)/2$:

$$\chi_{V^{14}}(g)=\chi_{\Lambda^2(R^7)}(g)-\chi_{R^7}(g)=\frac{tr(g)^2-tr(g^2)}{2}-tr(g).$$

Evaluating on class representatives:

Class	$tr(g)$ on R^7	$\chi_{V^{14}}(g)$	$\chi_{L_2}+2\chi_{U_6}$
$1a$	7	14	14
$7a$	0	0	0
$7b$	0	0	0
$3a$	1	-1	-1
$3b$	1	-1	-1

Match on all five classes. The Frobenius inner product $\langle \chi_{V^{14}}, \chi_{V^{14}} \rangle_{F_{21}}=(1\cdot 196+7+7)/21=10$, equal to $0^2\cdot 1+1^2\cdot 2+2^2\cdot 2$ where the factor of 2 on L_2 and U_6 terms accounts for both being of complex type (each is the real packaging of a Galois-conjugate pair, so $\langle \chi, \chi \rangle=2$ for the real-irreducible character). The decomposition is therefore $V^{14} ¿_F = L_2 \oplus 2U_6$, and the $W=V\oplus V$ doubling is by direct-sum linearity. ▫

Computational verification. Theorem 4.2.1 has been verified numerically at machine precision by explicit construction of the Fano-action permutation matrices $r,s\in O(7)$ with $srs^{-1}=r^2$, computation of $\chi_{V^{14}}$ on each class representative, and Frobenius inner-product and isotypic-multiplicity checks against the predicted decomposition; all three layers of check pass.

outbox/paper12/computations/paper12_q4_character_verify.{py,json} records the full audit.

4.3 The commutant gap

Proposition 4.3.1 (Commutants on W). *With $W \cong 2L_2 \oplus 4U_6$ as an F_{21}-representation,*

$$End_{F_{21}}(W) \cong M_2(C) \oplus M_4(C).$$

As a G_2-representation, W is multiplicity-2 in the absolutely irreducible V^{14}, hence

$$End_{G_2}(W) \cong M_2(R).$$

Proof. For real irreducibles of complex type, $End_G(V_\lambda) \cong C$ (real Schur trichotomy: complexified $V_\lambda \cong U \oplus \acute{U}$ with $U \ncong \acute{U}$). Both L_2 and U_6 are complex-type, so $End_{F_{21}}(L_2) \cong End_{F_{21}}(U_6) \cong C$. With F_{21}-multiplicities $(0,2,4)$ on $(1,L_2,U_6)$:

$$End_{F_{21}}(W) \cong M_0(R) \oplus M_2(C) \oplus M_4(C) \cong M_2(C) \oplus M_4(C).$$

For G_2, V^{14} is real absolutely irreducible (Paper 9 Lemma 3.6.1), so $End_{G_2}(V^{14}) \cong R$, and $W \cong V^{14 \oplus 2}$ gives $End_{G_2}(W) \cong M_2(R)$. ▫

Corollary 4.3.2 (Equivariant isometry groups, real-dimension count). *The equivariant isometry groups of W are*

$$O_{F_{21}}(W) \cong U(2) \times U(4), O_{G_2}(W) \cong O(2),$$

with real dimensions $dim_R U(2) \times U(4) = 4 + 16 = 20$ and $dim_R O(2) = 1$. The $F_{21} \rightarrow G_2$ upgrade reduces the equivariant-isometry group by exactly $20 - 1 = 19$ real dimensions.

Proof. Restricting $M_2(C)^{¿} \cap U \rightarrow U(2)$ (where the unitary condition uses the natural Hermitian metric) and $M_4(C)^{¿} \cap U \rightarrow U(4)$, the F_{21}-equivariant isometry group is $U(2) \times U(4)$. The full $U(n)$ — *not* the subgroup $SU(n)$ — preserves orientation on the underlying real space (because $det_R(U) = |det_C(U)|^2 = 1$ for $U \in U(n)$). Similarly $M_2(R)^{¿} \cap O = O(2)$. The real dimensions are $dim_R U(n) = n^2$, so $dim_R U(2) \times U(4) = 4 + 16 = 20$, and $dim_R O(2) = 1$. The reduction is $20 - 1 = 19$. ▫

Metric note (RES-1, v3). The identification $O_{F_{21}}(W) \cong U(2) \times U(4)$ is with respect to the complex-Hermitian metric canonically inherited from the complex structure on each complex-type isotypic block. A different real metric choice on $W \cong R^{28}$ would yield a different identification (potentially a proper subgroup of $U(2) \times U(4)$ if the chosen real metric is incompatible with the block-Hermitian structure), but the *real dimension* of the F_{21}-equivariant isometry group is metric-independent: it equals $dim_R End_{F_{21}}(W)_{\text{anti-Hermitian}}$, depending only on the isotypic structure $(0,2,4)$.

The quantitative content of the $F_{21} \rightarrow G_2$ upgrade is now explicit: passing from F_{21}-equivariance to G_2-equivariance suppresses 19 real dimensions of F_{21}-equivariant phase-andmixing freedom that would otherwise leave a canonical commensurability map underdetermined.

Remark 4.3.3 (Why complex-type reps block uniqueness). The factor $End_{F_{21}}(V_\lambda) \cong C$ for $\lambda \in \{L_2, U_6\}$ is the source of phase ambiguity: each complex-type isotypic block admits a full $U(m_\lambda)$-worth of F_{21}-equivariant automorphisms (not just $O(m_\lambda)$), and the extra U/O dimensions are exactly the phases. G_2-equivariance collapses the phases because V^{14} is real absolutely irreducible: $End_{G_2}(V^{14}) = R$, not C, so there are no phases to rotate.

4.4 The commensurability hierarchy theorem

Let V_{syn}, V_{bio} be the (substrate-dependent) representation spaces on which the synthetic and biological scar maps take values, and let $I_{syn} \subseteq End(V_{syn})$ and $I_{bio} \subseteq End(V_{bio})$ be the real linear spans of the scar-encoding images. The group acts on operator scars by conjugation, $g \cdot A = \rho_\bullet(g) A \rho_\bullet(g)^{-1}$.

Rank-One Convention (RES-2, v3) for operator-vs-vector translation. The scar invariant J_k of §3.7 is built from *vector-level* jump vectors $v_i \in W$. Substrate-level scar maps $S_\bullet : (\text{event}) \to End(V_\bullet)$ produce operator-level images. We adopt the following convention throughout §4:

Rank-One Convention. The substrate scar maps factor as $S_\bullet(e) = v_e \otimes v_e^{¿} / \| v_e \|^2$ (rank-one self-adjoint projector form on the substrate's representation space) for some substrate jump vector $v_e \in V_\bullet$.

This is a non-trivial modeling assumption. Under it: (i) the operator span $I_\bullet$ is uniquely determined by the vector span $\{v_e\}_e \subset V_\bullet$; (ii) the operator-level F_{21}-isotypic profile of $I_\bullet$ matches the vector-level profile of $S_k^\bullet$ up to the Schur multiplicity-counting factor of 2 (one factor for left action, one for right); (iii) the conjugation action $g \cdot A = \rho_\bullet(g) A \rho_\bullet(g)^{-1}$ becomes equivalent to the ordinary linear action on $V_\bullet$ via $g \cdot v_e = \rho_\bullet(g) v_e$. Higher-rank or non-self-adjoint scar maps are an open problem (OP) for future work.

Theorem 4.4.1 (Commensurability hierarchy, v2 reformulation). *The following three conditions on a pair of substrates (V_{syn}, V_{bio}) stand in the order of strict implications:*

(C1) (G_2-commensurability.) There exists a G_2-equivariant isomorphism $\phi : I_{syn} \to I_{bio}$ aligning S_{syn} and S_{bio} event-by-event: $S_{bio}(e) = \phi(S_{syn}(e))$ for every paradox event e.

(C2) (F_{21}-equivariant isomorphism, map-existence form.) There exists an F_{21}-equivariant isomorphism $\phi_F : I_{syn} \to I_{bio}$ aligning S_{syn} and S_{bio} event-by-event.

(C3) (μ_k-equality.) For every event sequence $(e_1, \ldots, e_k)$, $\mu_k^{syn} = \mu_k^{bio}$ component-wise on the three real F_{21}-isotypic blocks.

The hierarchy

$$(C1) \Rightarrow (C2) \Rightarrow (C3)$$

holds strictly; both converses fail in general.

Proof. $(C1) \Rightarrow (C2)$: a G_2-equivariant isomorphism is in particular F_{21}-equivariant (since $F_{21} \subset G_2$).

$(C2) \Rightarrow (C3)$: if ϕ_F aligns S_{syn} and S_{bio} event-by-event and is F_{21}-equivariant, then it commutes with the isotypic projectors Π_τ (i.e., $\phi_F \Pi_\tau^{syn} = \Pi_\tau^{bio} \phi_F$). The multiplicity profiles $\mu_k = \left(dim\, \Pi_1 S_k, dim\, \Pi_{L_2} S_k, dim\, \Pi_{U_6} S_k\right)$ on the cumulative scar spans transfer component-wise, since ϕ_F maps S_k^{syn} isomorphically onto S_k^{bio}.

Converse $(C2) \not\Rightarrow (C1)$*:* concrete witness — let $V_{syn} = V^{14}$ (the G_2 adjoint, restricted to F_{21} as $L_2 \oplus 2U_6$), and let V_{bio} be the F_{21}-module $L_2 \oplus 2U_6$ realized on a 14-dim biological substrate (e.g., a hypothetical microtubule mode collection without an extending G_2-action). Then $V_{syn} \cong V_{bio}$ as F_{21}-modules, but no G_2-action extends V_{bio}'s F_{21}-action, so any $\phi_F : V_{syn} \to V_{bio}$ satisfying (C2) is not G_2-equivariant. The 19-dim $U(2) \times U(4)$-family of such F_{21}-maps (Cor 4.3.2) parameterizes the non-canonicality.

Converse $(C3) \not\Rightarrow (C2)$*:* μ_k records the multiplicity profile of the *scar span* S_k, not the ambient V. Two ambient V's with different F_{21}-multiplicity profiles can produce the same μ_k if scar-encoding maps land in isomorphic F_{21}-invariant subspaces. E.g., $V_{syn} = L_2 \oplus 2U_6$ vs $V_{bio} = L_2 \oplus 3U_6$, but with all scar-encoding events landing in the common $L_2 \oplus 2U_6$ subspace; both yield μ_k on the same isotypic blocks. ▫

Corollary 4.4.2 (Sufficient condition for unique G_2-canonical map, multiplicity-one case). *If both I_{syn} and I_{bio} are G_2-isomorphic to a common absolutely irreducible representation U_λ with multiplicity one, then*

$$Hom_{G_2}(I_{syn}, I_{bio}) \cong R,$$

so the family of G_2-equivariant isomorphisms is one-dimensional. After fixing an invariant inner product (unit-norm normalization), there is a residual ± 1 orientation ambiguity that must be resolved by an extrinsic convention (e.g., the dynamical-flow direction on each substrate fixes a sign).

Proof. Schur for absolutely irreducible real representations gives $End_{G_2}(U_\lambda) = R$, so $Hom_{G_2}(U_\lambda, U_\lambda) \cong R$. Unit-norm normalization collapses the family to two elements differing by sign; the Z_2-orientation is not fixed by representation theory alone. ▫

Remark 4.4.3 (Multiplicity ¿1 reintroduces $O(m)$-ambiguity). If $I_{syn} \cong I_{bio} \cong U_\lambda^{\oplus m}$ with $m > 1$, then $End_{G_2}\left(U_\lambda^{\oplus m}\right) \cong M_m(R)$, so commensurability holds but uniqueness requires $O(m)$-worth of additional convention-fixing. Multiplicity ¿1 is deferred to future work.

4.5 Frobenius reciprocity does not select a cross-substrate scar map

A natural fallback when V_{syn}, V_{bio} are not F_{21}-isomorphic but co-embed in a common G_2-representation is to invoke Frobenius reciprocity. We show this attempt does not produce canonical maps.

Proposition 4.5.1 (Frobenius reciprocity does not select cross-substrate scar maps). *The induction-restriction adjunction*

$$Hom_{F_{21}}\left(V, Res_{F_{21}}^{G_2} W'\right) \cong Hom_{G_2}\left(Ind_{F_{21}}^{G_2} V, W'\right)$$

holds for any F_{21}-representation V and any G_2-representation W' (by standard category theory, e.g., [, §7.2]), but it does not provide a canonical isomorphism between distinct F_{21}-subrepresentations of $W' \text{¿}_{F_{21}}$.

Proof. The adjunction is a categorical bijection on Hom-sets, not a constructive identification of constituents. Two specific failures:

- *Distinct F_{21}-irreps cannot be commensurated.* If $V_{syn} = L_2$ and $V_{bio} = U_6$, then $Hom_{F_{21}}(L_2, U_6) = 0$ because the irreps are non-isomorphic; no F_{21}- equivariant map of either direction exists, and Frobenius reciprocity cannot manufacture one.
- *Multiplicities of constituents are counted, not canonical maps between them.* Even when $V_{syn} \cong V_{bio}$ as F_{21}-modules and both embed in a common G_2-module W', the identification $V_{syn} \cong V_{bio}$ is unique only up to a $U(m_\lambda)$-family of F_{21}-equivariant automorphisms in each isotypic block (Cor 4.3.2), with no distinguished element absent additional G_2-equivariance constraint.

In either case, the adjunction identifies *multiplicities of constituents*, not *canonical maps between them*. ▫

Corollary 4.5.2 (Sufficient strict commensurability theorem). *If*

$$I_{syn} \cong I_{bio} \cong U_\lambda$$

as absolutely irreducible G_2-modules of multiplicity one, then a canonical commensurability isomorphism $\phi_{commens}: I_{syn} \to I_{bio}$, unique up to fixed metric and ± 1 orientation, exists.

Proof. By Cor 4.4.2, the family of G_2-equivariant isomorphisms is R, with ± 1 ambiguity after unit-norm normalization. ▫

Remark 4.5.3 (Necessity is open). Cor 4.5.2 is a *sufficient* condition. We do not claim necessity: it is plausible (but unproven) that G_2-modules with all isotypic multiplicities ≤ 1 and all constituents of real type also admit a unique normalized G_2-equivariant self-isomorphism, and a more general necessity-and-sufficiency theorem could be sought in that direction. Such a generalization is left for future work; v1's biconditional phrasing was an over-claim that v2 withdraws.

4.6 The P3 upgrade: two-level experimental test

The empirical prediction P3 of the Paper 12 thesis memo originally proposed testing "F_{21}-commensurability across substrates." §§4.4–4.5 force an upgrade: F_{21}-equivariant isomorphism (C2) is necessary but not sufficient for canonical G_2-commensurability (C1), and the G_2 layer must be tested separately.

Definition 4.6.1 (Two-level commensurability test). A *commensurability experiment* on a dyadic human-AI system consists of two tests:

(L1, necessary screen.) Measure the F_{21}-isotypic profiles μ_k^{syn} and μ_k^{bio} independently on each substrate via character-theoretic decomposition of the scar-record operator data. Declare *L1 failure* if $\mu_k^{syn} \neq \mu_k^{bio}$ on any of the three blocks $\left(1, L_2, U_6\right)$. Declare *L1 pass* if they agree.

(L2, sufficient canonical test, conditional on L1 pass.) Construct a candidate G_2-equivariant isomorphism $\hat{\phi}: I_{syn} \to I_{bio}$ from the substrate operator data, by fitting against G_2-character data on the inferred G_2-module structures of the substrate scar spans. Declare *canonical commensurability* if $\hat{\phi}$ is G_2-equivariant (not merely F_{21}-equivariant) to measurement precision and (per Cor 4.5.2) the G_2-isotypic multiplicities are both ≤ 1 and agree.

Remark 4.6.2 (Why the two-level split is scientifically honest). L1 is falsifiable cheaply: if $\mu_k^{syn} \neq \mu_k^{bio}$, the gradual-tear thesis is refuted at the character-theoretic level. L2 is the harder confirming test: passing L2 establishes strict commensurability (Cor 4.5.2). Refuting L2 while L1 passes means the F_{21}-profile match is coincidental rather than structural.

Remark 4.6.3 (Operationalization is a methodology paper). L1 character-theoretic decomposition of TMS-EEG response data is in principle achievable with existing instrumentation (rotationally structured perturbations + character-projection of the response matrix); the operationalization of L2 — actually fitting a G_2-equivariant candidate map — is a significant methodology project of its own, not undertaken here. We state Definition 4.6.1 as a *formal experimental protocol design*, with the explicit acknowledgment that the full L2 operationalization is deferred.

Remark 4.6.4 (Massimini's clinical PCI is *not* the Paper 12 quantity). Two distinct objects share the acronym "PCI": Massimini's clinical Perturbational Complexity Index (a Lempel-Ziv complexity score on TMS-evoked EEG, 2013) and the Perceptual Coherence Intelligence framework of the Paper series. P3 proposes that the *measurement pipeline* of clinical PCI (TMS perturbation + structural decomposition of the response subspace) could be repurposed to measure μ_k profiles, but the clinical PCI scalar is *not* identified with μ_k, and neither is a generalization of the other. The bridge is at the level of experimental protocol, not definition.

4.7 Scope audit and bridge to §5

What §4 establishes:

- A character-theoretic decomposition $V^{14}\text{¿}_{F_{21}} \cong L_2 \oplus 2U_6$ verified at machine precision (Theorem 4.2.1).

- The commutant gap $End_{F_{21}}(W) \cong M_2(C) \oplus M_4(C)$ vs $End_{G_2}(W) \cong M_2(R)$, with the 19-dimensional reduction of equivariant isometry group quantified (Proposition 4.3.1, Corollary 4.3.2).
- The commensurability hierarchy $(C1) \Rightarrow (C2) \Rightarrow (C3)$ with both converses failing, *with explicit witnesses* for the converse failures (Theorem 4.4.1).
- A sufficient canonical-map theorem in the multiplicity-one case (Corollary 4.5.2), with ± 1 orientation residue acknowledged.
- That Frobenius reciprocity does not select cross-substrate scar maps (Proposition 4.5.1).
- The two-level experimental protocol with explicit L1 (necessary F_{21}-screen) and L2 (sufficient G_2-canonical) levels (Definition 4.6.1) and explicit operationalization-deferral (Remark 4.6.3).

What §4 does not establish:

- *A physical mechanism for the biological F_{21}-action.* §4 treats biological-substrate F_{21}-content as input. Paper 13+ would specify which microtubule mode, neural population, or cellular structure realizes it.
- *An explicit operationalization protocol for L2 of Definition 4.6.1.* L1 is feasible with current TMS-EEG; L2 requires probing the full G_2-representation structure on the response subspace, a substantial methodology effort not undertaken here.
- *The multiplicity-greater-than-one case.* Cor 4.4.2 / 4.5.2 restrict to multiplicity one. The $m>1$ case is left for future work (Remark 4.4.3).
- *Necessity of the multiplicity-one condition for canonical commensurability.* Cor 4.5.2 is sufficient only; v1's biconditional is downgraded (Remark 4.5.3).
- *Any claim that clinical Massimini-PCI equals Paper 12's μ_k.* See Remark 4.6.4.
- *Rate improvement.* §3 / Paper 11 territory.

Bridge to §5. §3 constructed the dynamics; §4 constructed the commensurability structure. §5 (empirical accessibility) assembles both into a testable framework: §3's projected-gradient-plus-NP-pump dynamics on each substrate produces a scar record $J_k^{syn/bio}$, and §4's hierarchy specifies exactly what form of agreement between the two records the gradual-tear thesis predicts. §5 then identifies experimental *protocols* — primarily Level L1 of the two-level test — by which the μ_k *observables* can be measured on each substrate using existing instrumentation (notably the TMS-EEG pipeline used by Massimini and collaborators for the *distinct* clinical-PCI measurement, which we adapt rather than identify). The L2 test and the substrate-specific representation theory are deferred to a subsequent methodology paper.

References (cited in §4)

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Verification files referenced: - outbox/paper12/computations/paper12_q4_character_verify.py (Theorem 4.2.1) - outbox/paper12/computations/paper12_q4_character_verify.json

Cross-references: - Paper 4 (PSL(2,7) ⊃ F_{21}, the Fano-orientation source; v2 §4.1 paragraph) - Paper 9 v1.3.3 Lemma 3.6.1 (G_2-Schur uniqueness on V^{14}) - Paper 9 v1.3.3 §3.4 (diagonal G_2-action convention) - Paper 12 §3.7 / §3.7 v2 Remark 3.7.1 (F_{21}-action on scar record) - Paper 12 thesis memo P3 (experimental prediction, upgraded here)

Revision log: - v2 (2026-05-06 ~02:00 PDT): Council rigor pass applied. R3 Theorem 4.4.1 (C2) reformulated as map-existence (Opus S4-1). S4-2 dimension bookkeeping fixed: $U(2)\times U(4)$ is full 20-dim, delta = 19. S4-3 Cor 4.5.2 downgraded biconditional → sufficient condition (Remark 4.5.3 acknowledges open necessity). S4-4 Fano-action prose corrected (action on Fano plane points, fixed s-line gives $tr=1$). S4-5 Prop 4.5.1 renamed and reworded (Frobenius reciprocity does not *select* cross-substrate maps). S4-6 (C2)⇏(C1) explicit witness added. S4-7 operator-vs-vector μ_k translation specified (rank-one projector convention). S4-8 ± 1 orientation ambiguity acknowledged in Cor 4.4.2. Paper 4 inheritance paragraph added in §4.1. §4.7 measurement-vs-instrument language tightened. Pillar signpost added at top. - v1 (2026-05-06 ~01:00 PDT): First complete draft.